

How can cloud computing change the future of climate action?

Introduction

Climate change is a common topic in the modern world, and one of the main questions is how to collectively overcome the impacts of climate change. This is the target of the UN SDG 13. Understanding how to mitigate the impacts has increased importance significantly: constantly we see examples of extreme or abnormal weather behaviour (Levin, Beisekenov et al.) there is a need for an innovative solution, and the need is becoming increasingly urgent as time passes. As the threat of climate change is becoming greater than ever before.

The need for advanced technological approaches based on big data is essential, to adapt and enhance resilience in the battle against climate change. More recent examples of developments in cloud computing and the benefit that it brings regarding UN SDG 13 (Climate Action). Efforts can be seen; as the development of GEE (Google Earth Engines) – a cloud computing based platform. The advantage of GEE is its remarkable computation speed (Udovychenko, 2022) allowing for large scale analysis of the effects of climate action VIA satellite imagery.

The evolution of cloud computing

Cloud Computing is the modern form of computing, basing its foundations around an ondemand service, giving users access to a wider range of resources. Cloud computing is often formed on a subscription basis or pay as you need basis, meaning that they are easily scalable for user's diverse needs. Furthermore, cloud-based computing is a non-local service. Meaning they are usually large-scale off-site servers and computers systems all managed by the external cloud service providers, often pooling / sharing these resources with multiple additional users (Manvi and Shyam, 2014.). Cloud computing, described as an umbrella term to refer to a large scale sophisticated on demand computing services (Voorsluys, W., Broberg, J et al). Cloud computing is often referred to as The Fifth Generation of Computing (after Mainframe, Personal Computer, Client-Server, Computing, and the web). (Padhy and Patra, 2012).

Cloud computing services are often categorised into three distinctive areas/ models these areas are: platform as a service (PAAS), infrastructure as a service (IAAS) and software as a service (SAAS).

Cloud computing allows for the move away from localised hardware management and rather the conversion to cloud based computing services, one of the key models that has allowed us to facilitate this change is the IAAS model (infrastructure as a service). IAAS has allowed for cloud based computing providers to provide consumers with on demand services such as VMS, Storage, and networking. (Manvi and Shyam, 2014.) With the context of UN SDG 13. IAAS allows for scalable computing capacities and large-scale storage that is required to host, manage and maintain high resolution data imagery such as satellite imagery and geographical archives for future comparisons to help with predictive analysis of future climate events.

PAAS builds onto the model of IAAS- Platform as a service is a virtualized platform that consists of many servers. It is formed of layers of software's and provides it as a service that can be used to make more complex services(Singh and Sharma et al.)Users are able to use platforms of high complexity through the cloud service provider, in short summary PAAS has preconfigured development tools to offer for users of cloud computing which allows for companies to allocate time/ or gain access to high specialised environments and software, without managing the hardware. PAAS may be beneficial for scenarios that may require high data processing power and complex modelling.

The final model that interlinks with both PAAS and IAAS – Software as a Service (SaaS) is a newly emerging business model used in the software industry (Satyanarayana, 2012). SaaS provides businesses and customers with the software that they would get with in business implementation of software. SaaS is seen as the final aspect – the delivery layer for the final application developed using IAAS and PAAS that end users can interact with in and amongst a company, such as a dashboard which alerts users of wildfire risks, or large-scale flooding risks. This layered approach allows researchers to bypass traditional hardware limitations and focus on planetary-scale geospatial analysis.

Technological implications

The primary implication of cloud computing is not just the storing large amounts of data or the processing of data, rather it is that there aren't enough systems that carry out both, the scale of the environment but also the specialisation of environments to carry out these studies; a prime example of a platform that does both of these using cloud computing services which benefits the UN sustainable development goal 13 (climate action) is GEE(Google Earth Engine). The problem is that scientists must manually search for and download multiple petabytes (1000 terabytes) of satellite imagery data, which can take days and weeks to analyse a small percentage of the substantial amounts of satellite data. Addressing big data is challenging and time-consuming task that requires a large computational infrastructure to ensure successful data processing and analysis (Hashem, et al) this is where GEE is introduced.

GEE inverts this implication, GEE is an example of a cloud based platform that uses all of these models provided by cloud computing it uses large amounts of computational power which analyses 40+ years of geospatial data / satellite imagery to come up with accurate predictions of climate change, which scientists and researchers could use to simulate climate events and how to prevent and mitigate these events. An example of this may be flooding, deforestation wildfire protection: with the large amounts of geospatial data due to cloud computational abilities, GEE is able to map out flooding and identify what areas may be of risk of flooding based on differences over the years, which allows for measures to be put in place to avoid and mitigate these risks GEE is described as in another study as a (Gorelick, N., Hancher et al) "cloud-based platform for planetary-scale geospatial analysis" gathering Google's massive computational capabilities that have variety of high-impact societal issues including deforestation, drought, disaster, disease, food security, water management, climate monitoring and environmental protection tools

Legal concerns

Legal concerns that may arise with cloud computing is that the infrastructure, data, and the service is not owned by the user; it is owned by the cloud service provider, this leads to legal questions about who the data belongs to. There are three main areas that this issue of who the owner of the data is, the provider can usually be classed as one of three therefore creating a grey area: legal custodian, owner or possessor (Krishnan, and Chen), cloud computing companies may have different levels of framework regarding data security, e.g./ Europe must follow the GDPR. some cloud computing companies also have access to the data which may be a breach of data privacy if data stored is confidential. In the context of climate action, this creates a conflict of 'Data Sovereignty': if a private company like google owns the data, do they have the right to withhold vital flood warnings/ deforestation data from the government. This highlights a key issue: 'Data sovereignty.'

Ethical concerns

With cloud computing additionally there are ethical concerns regarding data privacy and confidentiality- with cloud computing users of the service user the visibility of how and where the data is being stored by third party cloud computing services and who has access to the data. This creates the 'black box' problem, if a climate model is hacked/ biased it could lead to false alarms or missed warnings – which could lead to the loss of life. For climate scientists, the lack of visibility is critical. There might not be clear evidence that your data has been accessed by an external party or not (Faragardi, 2017) this also adds to the concern regarding security and the integrity of cloud hosted data. undermining global trusts in climate models

Conclusion

In conclusion, the integration of cloud computing into climate strategy represents an increasing fundamental shift in the approach to tackle the climate crisis, with technological developments, technology has been able to evolve from complex models such as IAAS and PAAS to provide large scale planetary scale data handling. As demonstrated by GEE, the rapid ability to process petabytes of satellite imagery due to cloud computing allows for increasing chance to mitigate climate disasters such as flooding and deforestation. Directly supporting the UN sustainable development goal 13 (Climate action).

Despite this, with cloud computing as discussed there are still grey areas causing friction, as previously identified regarding ownership of data and third-party external access, the legal and ethical implications that are posed by cloud computing despite its wonders cannot be ignored. As cloud computing moves forwards to resolve this issue, changes will need to be made, for us to be able to fully leverage the power of cloud computing, by implementing a standardised worldwide framework of data handling. Enabling for the use cloud computing as a large-scale platform to help combat climate change rather a source of legal and ethical vulnerability.

Word count: 1499

References

- Faragardi, H.R., 2017, June. Ethical considerations in cloud computing systems. In *Proceedings* (Vol. 1, No. 3, p. 166). MDPI. [Accessed 20/11/2025]
- Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D. and Moore, R., 2017. Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote sensing of Environment*, 202, pp.18-27. [Accessed 15/11/2025]
- Hashem, I.A.T., Yaqoob, I., Anuar, N.B., Mokhtar, S., Gani, A. and Khan, S.U., 2015. The rise of “big data” on cloud computing: Review and open research issues. *Information systems*, 47, pp.98-115. [Accessed 04/12/2025]
- Krishnan, S. and Chen, L., 2019. Legal concerns and challenges in cloud computing. *arXiv preprint arXiv:1905.10868*. [Accessed 25/11/2025]
- Levin, E., Beisekenov, N., Wilson, M., Sadenova, M., Nabaweesi, R. and Nguyen, L., 2023. *Empowering climate resilience: Leveraging cloud computing and big data for community Climate Change Impact Service (C3IS)*. *Remote Sensing*, 15(21), p.5160.[Accessed 01/11/2025]
- Manvi, S.S. and Shyam, G.K., 2014. Resource management for Infrastructure as a Service (IaaS) in cloud computing: A survey. *Journal of network and computer applications*, 41, pp.424-440. [Accessed 13/11/2025]
- Padhy, R.P. and Patra, M.R., 2012. Evolution of cloud computing and enabling technologies. *International Journal of Cloud Computing and Services Science*, 1(4), p.182.[Accessed 12/11/2025]
- Satyanarayana, S., 2012. Cloud computing: SAAS. *Computer Sciences and Telecommunications*, (4), pp.76-79. [Accessed 03/12/2025]
- Singh, A., Sharma, S., Kumar, S.R. and Yadav, S.A., 2016, February. Overview of PaaS and SaaS and its application in cloud computing. In *2016 International Conference on Innovation and Challenges in Cyber Security (ICICCS-INBUSH)* (pp. 172-176). IEEE. [Accessed 04/12/2025]
- Udovychenko, V., 2022, November. *Monitoring the Nexus Between Forest Loss, Land Cover Dynamics, and Climatic Changes by GEE GIS Facilities for the Landscape Planning Implementation*. In *16th International Conference Monitoring of Geological Processes and Ecological Condition of the Environment* (Vol. 2022, No. 1, pp. 1-5). European Association of Geoscientists & Engineers. [Accessed 18/11/2025]
- Voorsluys, W., Broberg, J. and Buyya, R., 2011. Introduction to cloud computing. *Cloud computing: Principles and paradigms*, pp.1-41. [Accessed 04/12/2025]

